When the total traffic volume is the same, the factor(s) which affect(s) the noise pollution level is/are:

(a) % of heavy vehicles
(b) Noise pollution level and average noise level
(c) Average noise level and % of heavy vehicles
(d) Indeterminable on the basis of details given

Ans. (a) : Considering the table and details of traffic volume per for location and when the volume of total traffic is the same then the percentage of heavy vehicle affects the noise pollution level.

146. The price fluctuations of 4 scrips in a stock market in the four quarters of a year are shown in the table below. Four different investors had the following portfolios of investment in the four companies throughout the year:

Portfolios
Investor 1 : 10 of A, 20 of B, 30 of C and 40 of D
Investor 2 : 40 of A, 10 of B, 20 of C and 30 of D
Investor 3 : 30 of A, 40 of B, 10 of C and 20 of D
Investor 4 : 20 of A, 30 of B, 40 of C and 10 of D

Stock Market Performance

	I Quarter	II Quarter	III Quarter	IV Quarter
Scrip A	Up 10%	Down 15%	Up 10%	Down 10%
Scrip B	Up 2%	Up 1%	Up 2%	Up 2%
Scrip C	Up 1%	Up 1%	Down 5%	Down 1%
Scrip D	Up 20%	Down 15%	Up 30%	Down 10%

In the light of the above which one of the following statements is correct?

(a) Investor 2 has made the best investment
(b) Investor 1 has made the best investment
(c) Investor 2 suffered a net loss during the year
(d) Investor 3 suffered a net loss during the year

Ans. (b) : Analyzing the investor's portfolios and calculating with stock market performance investor has made the best investment.

147. X and Y are two variables whose values at any time are related to each other as shown in Fig. (i). X is known to vary periodically with reference to time as shown in Fig. (ii).

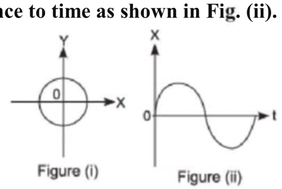
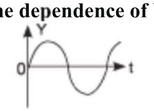


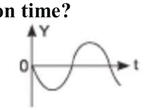
Figure (i)

Figure (ii)

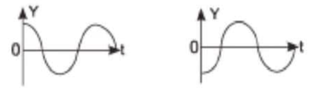
Which of the following curves depicts correctly the dependence of Y on time?



(a)



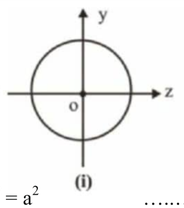
(b)



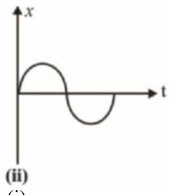
(c)

(d)

Ans. (c) :

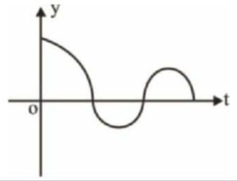


(i)



(ii)

(i) $x^2 + y^2 = a^2$ .....(i)
(ii) $x = a \sin \omega t$ .....(ii)
So $y^2 = a^2 - x^2$
$= a^2 - (a \sin \omega t)^2$
$= a^2 - a^2 \sin^2 \omega t$
$= a^2 (1 - \sin^2 \omega t)$
$y^2 = a^2 \cos^2 \omega t$
$y = \pm a \cos \omega t \rightarrow$ which is shown as.



148. Out of a total of 120 musicians in a club, 5% can play all the three instruments-guitar, violin and flute. It so happens that the number of musicians who can play any two and only two of the above instruments is 30. The number of musicians who can play the guitar alone is 40. What is the total number of those who can play violin alone or flute alone?

(a) 45 (b) 44
(c) 38 (d) 30

Ans. (b) : Total musicians in a club is 120
5% of all can play all the three instrument
$= \frac{5}{100} \times 120 = 6$ member
Who can play any two and only two of the above instrument is 30.
40 person can play guitar alone
Let y = 30, z = 40
So, those who can play violin alone or flute alone.
$= 120 - (x + y + z) = 120 - (6 + 30 + 40) = 120 - 76$
$= 44$

149. Which of the following can be inferred from the statement that 'Either John is stupid or John is lazy'?

I. John is lazy/therefore, John is not stupid.
II. John is not lazy/therefore, John is stupid.
III. John is not stupid/therefore, John is lazy.
IV. John is stupid/therefore, John is not lazy.
Choose the correct answer from the codes given below:
Codes:

(a) I and II (b) II and III
(c) III and IV (d) I and IV

Ans. (b) : As per the statement, John is either lazy or stupid. But he cannot be both at the same time.
So, only II and III are correct.

150. A person earns Rs. 2000 per month over and above his salary as additional charge allowance. However, 30% of this additional income will be deducted as additional income-tax at source. If the person would deposit Rs. 1000 per month on a long-term saving fetching 12% interest his tax liability on the additional allowance would reduce to 10%. What is the effective interest for this person for money invested in the long-term savings scheme?

(a) 12% (b) 18%
(c) 19% (d) 20%

Ans. (b) : The additional charge allowance is Rs. 2000/month
Additional Income Tax is 30% of this additional income
$= \frac{30}{100} \times 2000 = 600$ rupees
If person deposits Rs. 1000 per month on a long term additional allowance would reduce to 10%
So tax deduction on 600 rupees is = 10%
$= \frac{10}{100} \times 600 = 60$ rupees
The total gain will be 12% + 60 rupees
So, 12% of 1000 + 60 rupees $= \frac{12}{100} \times 1000 + 60$ rupees
So, total effective interest is 180 rupees on 1000 which is 18%.